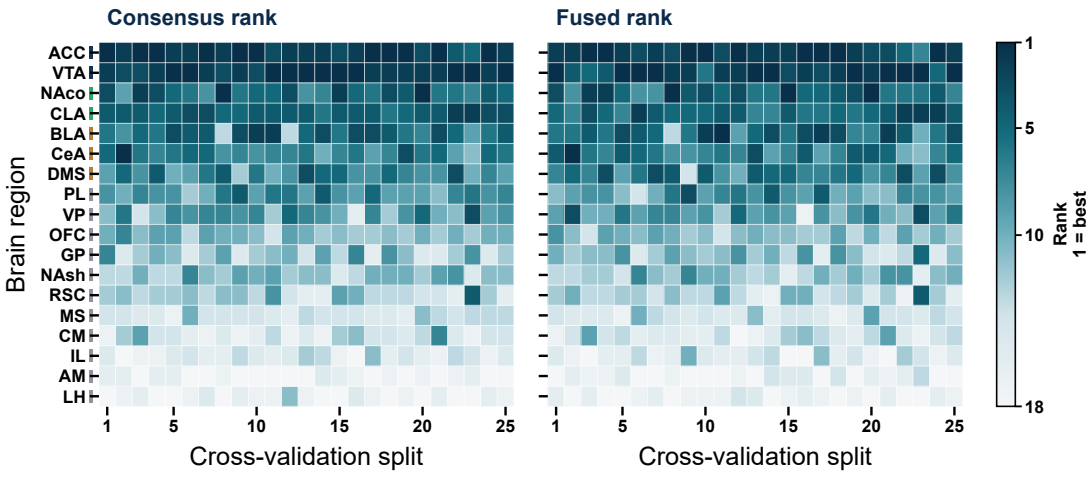
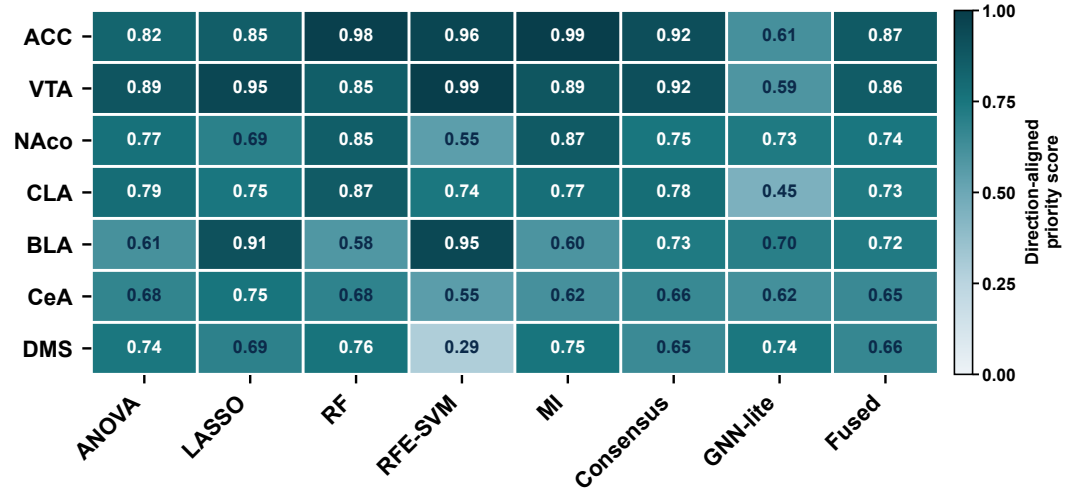
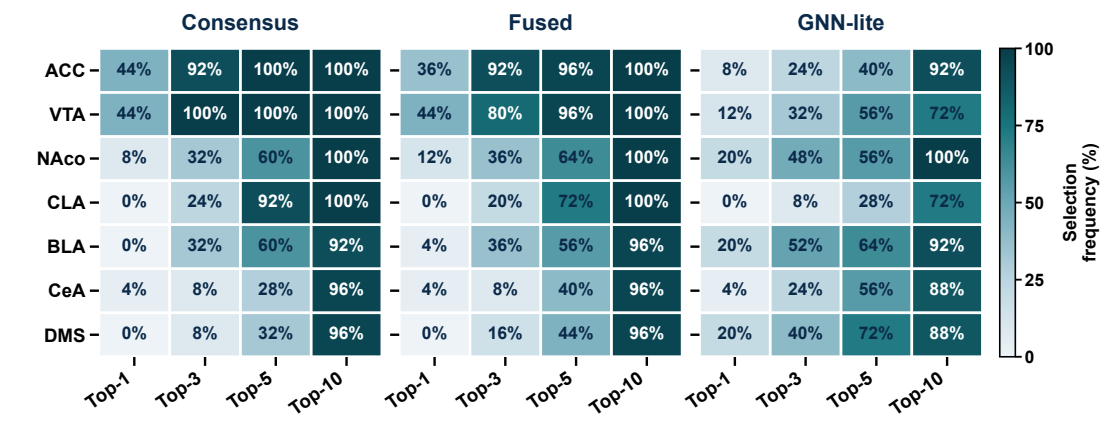
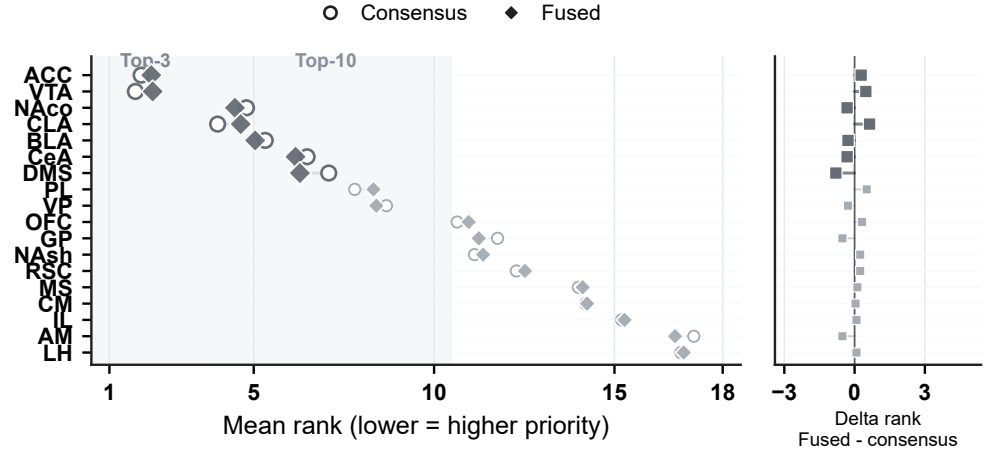
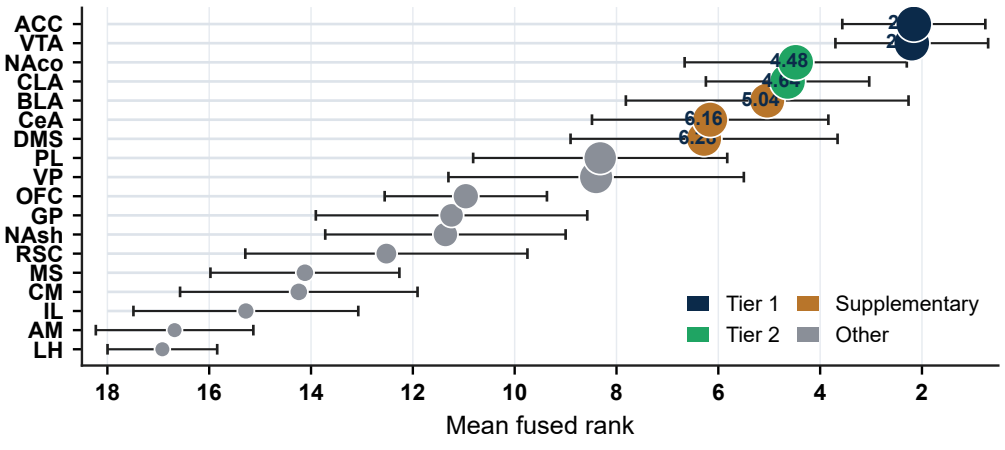
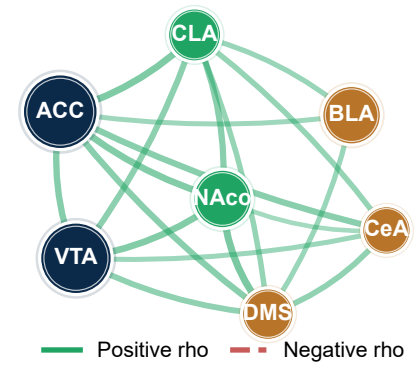

Consensus rank
Fused rank
Brain region
ACC
VTA
NAco
CLA
BLA
CeA
DMS
PL
VP
OFC
GP
NAsh
RSC
MS
CM
IL
AM
LH
1
5
10
15
20
25
Cross-validation split
Rank
1 = best
18

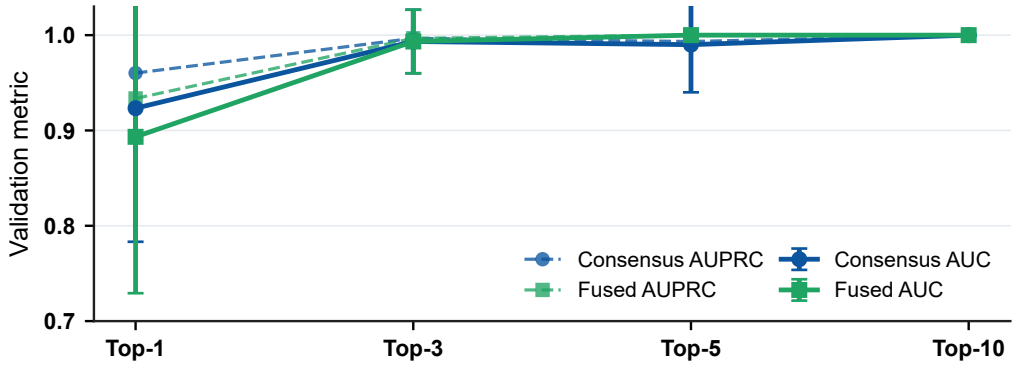

ACC 0.82 0.85 0.98 0.96 0.99 0.92 0.61 0.87
VTA 0.89 0.95 0.85 0.99 0.89 0.92 0.59 0.86
NAco 0.77 0.69 0.85 0.55 0.87 0.75 0.73 0.74
CLA 0.79 0.75 0.87 0.74 0.77 0.78 0.45 0.73
BLA 0.61 0.91 0.58 0.95 0.60 0.73 0.70 0.72
CeA 0.68 0.75 0.68 0.55 0.62 0.66 0.62 0.65
DMS 0.74 0.69 0.76 0.29 0.75 0.65 0.74 0.66
ANOVA
LASSO
RF
RFE-SVM
MI
Consensus
GNN-lite
Fused
Direction-aligned priority score
1.00
0.75
0.50
0.25
0.00

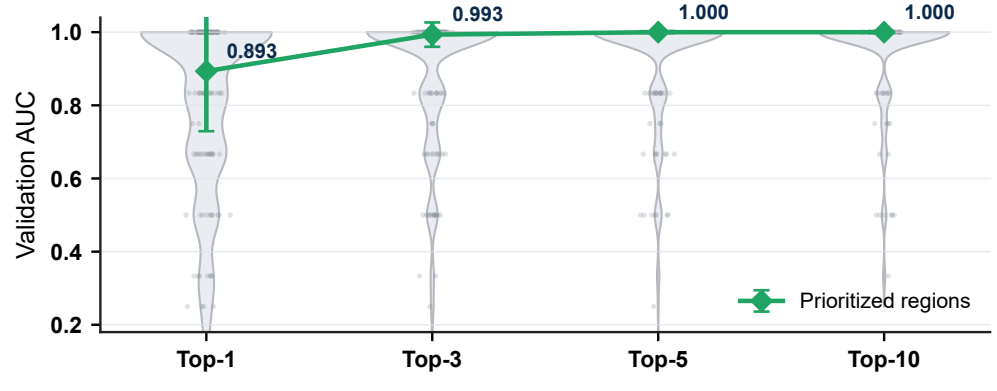

Consensus
Fused
GNN-lite
ACC 44% 92% 100% 100% 36% 92% 96% 100% 8% 24% 40% 92%
VTA 44% 100% 100% 100% 44% 80% 96% 100% 12% 32% 56% 72%
NAco 8% 32% 60% 100% 12% 36% 64% 100% 20% 48% 56% 100%
CLA 0% 24% 92% 100% 0% 20% 72% 100% 0% 8% 28% 72%
BLA 0% 32% 60% 92% 4% 36% 56% 96% 20% 52% 64% 92%
CeA 4% 8% 28% 96% 4% 8% 40% 96% 4% 24% 56% 88%
DMS 0% 8% 32% 96% 0% 16% 44% 96% 20% 40% 72% 88%
Top-1
Top-3
Top-5
Top-10
Selection frequency (%)
100
75
50
25
0


Consensus
Fused
Top-3
Top-10
ACC
VTA
NAco
CLA
BLA
CeA
DMS
PL
VP
OFC
GP
NAsh
RSC
MS
CM
IL
AM
LH
1
5
10
15
18
Mean rank (lower = higher priority)
−3
0
3
Delta rank
Fused - consensus


ACC
VTA
NAco
CLA
BLA
CeA
DMS
PL
VP
OFC
GP
NAsh
RSC
MS
CM
IL
AM
LH
4.48
5.04
6.16
Tier 1
Tier 2
Supplementary
Other
18
16
14
12
10
8
6
4
2
Mean fused rank


CLA
ACC
BLA
NAco
CeA
VTA
DMS
Positive rho
Negative rho


Validation metric
1.0
0.9
0.8
0.7
Consensus AUPRC
Consensus AUC
Fused AUPRC
Fused AUC
Top-1
Top-3
Top-5
Top-10


Validation AUC
1.0
0.8
0.6
0.4
0.2
0.893
0.993
1.000
1.000
Prioritized regions
Top-1
Top-3
Top-5
Top-10